

Information-optimal illumination for dead-time-limited single-pixel imaging: certificates, water filling, and the jitter-capped ridge

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Abstract

A photon-counting detector operated at its deterministic information-optimal flux can forfeit more than half of its recorded information the moment its dead time carries a few percent of timing jitter. This paper asks how hard a real, jittery detector can be driven, and with what illumination — one law answers the first half, one certified design principle the second, and a preregistered campaign seals both. The law: for hidden random hold times of coefficient of variation c_v , we derive an exact missing-information identity and the long-window count-information rate $J_\infty(\rho, c_v) = \rho/[(1+\rho)(1+c_v^2\rho^2)]$, whose optimum is finite for every nonzero hold-time variance, solves the cubic $c_v^2\rho^2(1+2\rho) = 1$, and scales as $2^{-1/3}c_v^{-2/3}$ at small jitter; matching this extensive loss to the deterministic terminal-phase loss of a companion study [?] predicts the smooth crossover $\rho^*(\nu, c_v) \sim (6\nu)^{1/3}(1+12\nu c_v^2)^{-1/3}$, with zero jitter the critical manifold on which the ridge grows as $\nu^{1/3}$ ($\nu = T/\tau$ dead-time slots per exposure). Exact-law peak predictions registered before the refined Monte-Carlo sweep completed were confirmed out-of-sample at both fresh points (3% and 6%), and a pooled two-run continuous-peak fit measures the jitter exponent at -0.658 (1.3% from $-2/3$). The design principle: hand-designed sparse and matched patterns are atoms of one concave experiment-design program whose per-atom information is the exact renewal kernel, and under deployment safety that program yields a sharply localized doctrine. Information rewards scene-adapted measurement geometry; a uniform $\pm 5\%$ per-pixel dose constraint controls cumulative exposure but does not eliminate that geometry headroom — a development-only constructive probe exhibits a dose-uniform, budget-feasible design with more than twice the local D-information of the balanced family in audited development cells; whether the headroom survives the full deployed conditioning stack — incident budget, dose band, A-risk cap, and spectral floor — is tested by a 480-cell descriptive full-stack analysis of the balanced (SCAT32) design; and global operating-point control is orthogonal and survives, one power knob that tracks the jitter-capped ridge. A pre-

registered simulation campaign (M1) tests the doctrine with a paired fixed-dwell ridge operating-point primary (RIDGE-SCAT32 versus SCAT32-060, a power-for-time comparison), a nine-dwell time-to-quality secondary, the full-stack descriptive analysis, five arms, and a mandatory disclosure ledger. Using the preregistered elapsed optical-time coordinate, $T_{\text{opt}} = M_{\text{total}}\nu\tau$, and the frozen Q90/PAVA/censoring procedure, the ridge time-to-quality endpoint passed: the median conservative speed ratio was $19.13\times$, the family-stratified nested-bootstrap 2.5th-percentile bound was $18.33\times$, and 21 of 24 scenes had $S_{\text{gate}} > 1$. The ridge operating-point primary also passes (median fixed-dwell radiometric gain 1.87 dB at $\nu = 2000$, 19/24 scenes positive, 2.5% bootstrap lower bound 1.41348975 dB). The full-stack computation is descriptive: across 480 cells, none was certified, 299 produced feasible counterexamples, and 181 were numerically unresolved. The result is an operating handbook: measure the jitter, let the law set the load, read the certificate as a geometry diagnostic, and tune the global operating point when time is the scarce resource.

1 Introduction

Drive a photon-counting detector to the flux where its count information peaks, and a five-percent timing jitter in its dead time silently forfeits about 55% of that information (Table ??). Single-pixel imaging (SPI) and computational ghost imaging reconstruct a scene from the responses of a single bucket detector to structured illumination [?, ?, ?]; in the photon-starved regime the bucket is a photon-counting device that is blind for a dead time τ after each registered count, and the conventional response is to operate at low flux, keeping the per-slot load $\rho = \lambda\tau$ at the few-percent level and paying in acquisition time. A companion study [?] showed that this convention leaves gains on the table: the exact Fisher information of the integrated nonparalyzable count possesses a principal ridge at the cube-root load $\rho^*(\nu) = (6\nu)^{1/3} - 2/3 + O(\nu^{-1/3})$, and that work mapped when driving a fixed illumination family toward the ridge helps. The trap above is the sharp edge

of that map: the deterministic ridge is not where a *real* detector should operate. Figure 1 sets out the dead-time-aware imaging pipeline, the global power knob, and the count-information ridge law that anchors the paper.

One law answers the deployment question: hidden hold-time randomness adds an *extensive* missing-information term, the long-window optimum becomes finite at the root of an exact cubic, and the deterministic ridge is the zero-jitter critical manifold of a two-parameter crossover (Section 2); its peak predictions, registered before the refined Monte-Carlo sweep existed, were confirmed out-of-sample. One certified design principle answers the design question: hand-designed patterns are atoms of a single concave experiment-design program over design measures with the exact renewal kernel; uniform dose alone controls cumulative exposure without eliminating scene-adapted geometry headroom, so the operative restriction is the full deployed conditioning stack, against which the balanced design’s near-optimality is tested by a certified conic dual bound over an expanded frozen dictionary, with the global detector operating point as the orthogonal surviving control (Section ??). A preregistered campaign (M1) seals both with a ridge operating-point primary, a time-to-quality secondary, the full-stack certification, and a mandatory disclosure ledger (Sections 3–4).

We formulate a local approximate-design framework for single-pixel illumination in which each pattern’s information is computed from the exact finite-window nonparalyzable renewal count law, optimized under explicit detector-load, dose, per-pixel-exposure, and peak-irradiance constraints and accompanied by generalized-equivalence and exact-rounding certificates. Existing photon-counting optimal experiment design, adaptive or learned SPI, and dead-time optimal-flux studies address neighboring components [?, ?, ?, ?, ?, ?]; we found no prior framework combining these elements for integrated-count single-pixel illumination. The certificate is always stated relative to the frozen dictionary, the local pre-scan estimate, and the declared resource model.

This work contributes: (i) the random-hold count-information law (Section 2): an exact missing-information identity, the long-window rate with its exact jitter-limited cubic optimum and small-jitter coefficient $2^{-1/3}$, and the matched crossover prediction $(6\nu)^{1/3}(1 + 12\nu c_v^2)^{-1/3}$, verified out-of-sample against preregistered predictions with a pooled jitter-exponent measurement of -0.658 ; (ii) the certified design principle (Section ??): one concave program with the exact renewal kernel, a constructive development-stage localization showing that uniform dose alone leaves scene-adapted geometry headroom, a full-stack conic near-optimality certificate over an expanded gain-coupled

dictionary, frozen guards, and the photon-budget versus time-budget conjugacy as its operating doctrine; (iii) a preregistered simulation campaign (M1) whose scenes, atoms, load grid, guards, endpoints, and three mutually non-rescuing verdicts were frozen in an immutable commit before any confirmatory reconstruction existed and are reported as frozen: the ridge operating-point primary passes, the elapsed-optical-time speed endpoint passes, and the full-stack computation is a descriptive analysis returning counterexamples rather than certification; and (iv) an operating handbook for real detectors (Table 4), ending in a bench-testable prediction. All code, frozen preregistration documents, and per-cell evidence bundles are openly released; full derivations, guard machinery, and verification detail are in the supplement.

2 The law: driving a jittery detector

The recovery-time law of the detector determines a scalar count-information kernel; everything else in this paper allocates illumination against that kernel. The companion’s deterministic detector [?] is the zero-jitter critical manifold of the theory below. The supplement gives the installed proofs with their stated regularity and conditionality assumptions; the crossover law remains a matched-asymptotic prediction rather than a universal finite-window theorem.

2.1 Random-hold count information

Generalize the detector so that after each registered event it blocks for a hidden *random* hold time $B_i \sim H$, iid, independent of the Poisson arrivals and of λ , with mean τ and coefficient of variation $c_v = \sigma_B/\tau$ (the companion’s deterministic model is $c_v = 0$). The observed datum remains the integrated registered count N_T over $T = \nu\tau$; the active time L_T , the state path, and the realized holds are hidden from the count-only observer.

Theorem 1 (missing-information identity) *For $\theta = \log \lambda$, the complete detector-path information is $\mathbb{E}[N_T]$, and the information retained by the scalar count is exactly*

$$I_N(\theta; T) = \mathbb{E}[N_T] - \lambda^2 \mathbb{E}[\text{Var}(L_T | N_T)] \quad (1)$$

for every finite T and every iid hidden hold law H independent of λ .

The companion’s terminal-phase identity is the special case in which the uncertainty in L_T is confined to the terminal detector phase. The identity requires amendment if the hold law depends on λ or the scene, if realized holds are observed and used, if holds are serially correlated, or if the counter is paralyzable.

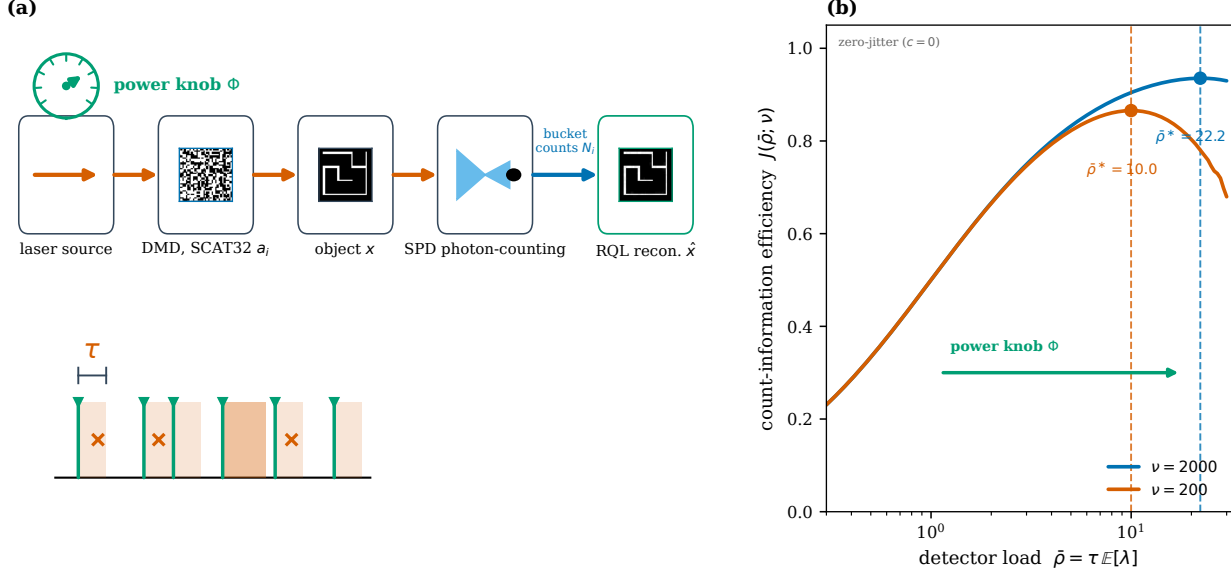


Figure 1: Schematic of the dead-time-aware single-pixel imaging pipeline and the count-information ridge law. (a) Imaging pipeline: the global power knob Φ (dial) is the single scalar source multiplier setting the detector load $\bar{\rho} = \tau \mathbb{E}[\lambda]$; the beam (orange arrows) passes from the laser source through the DMD, which displays the SCAT32 scattered-sparse illumination patterns a_i (inset), to the object x (inset: the confirmatory maze scene), and the SPD photon-counting detector feeds the bucket counts N_i (blue arrow) to the RQL reconstruction $\hat{x}$ (inset: the RIDGE-SCAT32 reconstruction of the same scene, seed 0, $\nu = 2000$). DMD: digital micromirror device; SPD: single-pixel (bucket) detector; RQL: renewal quasi-likelihood reconstruction. In the photon-arrival timeline (lower inset), recorded counts (green markers) each open a non-paralyzable dead-time hold of length τ (shaded; one hold is jitter-broadened, darker), and arrivals inside a hold are lost ($\times$); the exposure is $T = \nu\tau$, so $\nu = T/\tau$ is the dwell budget. (b) Count-information efficiency $J(\bar{\rho}; \nu)$ versus detector load, evaluated from the exact finite-window count law for $\nu = 2000$ and $\nu = 200$, whose optimal-load ridges are $\bar{\rho}^* = 22.2$ and $\bar{\rho}^* = 10.0$, respectively; the ridge follows the finite-window law $\bar{\rho}^*(\nu) = (6\nu)^{1/3} - 2/3$ (zero-jitter limit, $c = 0$), and the global power knob (arrow) moves the operating point onto the ridge. The colour is used only for display purposes.

Theorem 2 (long-window rate) Under standard whose leading coefficient is $2^{-1/3} = 0.79370\dots$ nonlattice renewal local-limit conditions,

$$\frac{I_N(\log \lambda; T)}{\nu} \longrightarrow J_\infty(\rho, c_v) = \frac{\rho}{(1 + \rho)(1 + c_v^2 \rho^2)}, \quad (2)$$

depending on H only through its mean and variance. The missing-information rate converges to $c_v^2 \rho^3 / [(1 + \rho)(1 + c_v^2 \rho^2)]$: an extensive loss equal, to first order in $c_v^2 \rho^2$, to the heuristic $\lambda^2 \sigma_B^2 \mathbb{E}[N_T]$ — conditioning on the count partially reveals the accumulated hold fluctuation, contributing the factor $(1 + c_v^2 \rho^2)^{-1}$ — and, unlike the deterministic terminal-phase loss, nonvanishing as $\nu \rightarrow \infty$.

Theorem 3 (exact jitter-limited optimum) For every fixed $c_v > 0$, $J_\infty(\rho, c_v)$ has a unique interior maximum ρ_c solving the cubic

$$c_v^2 \rho_c^2 (1 + 2\rho_c) = 1, \quad (3)$$

with small-jitter expansion

$$\rho_c = 2^{-1/3} c_v^{-2/3} - \frac{1}{6} + \frac{2^{1/3}}{36} c_v^{2/3} + O(c_v^{4/3}), \quad (4)$$

For exponential holds ($c_v = 1$, outside the small-jitter regime) the exact positive root is $\rho_c = 0.6573\dots$; gamma, bounded two-point, and lognormal holds sharing a small c_v share the leading coefficient, with distribution shape entering at the next order.

2.2 The crossover and its universality class

Matching the extensive loss $c_v^2 \rho^2$ against the deterministic terminal-boundary loss $\rho^2/(12\nu)$ in the joint regime $\nu \rightarrow \infty$, $c_v \rightarrow 0$, $c_v^2 \nu = O(1)$ gives

$$\rho_*(\nu, c_v) \sim \left(2c_v^2 + \frac{1}{6\nu}\right)^{-1/3} = (6\nu)^{1/3} (1 + 12\nu c_v^2)^{-1/3}, \quad (5)$$

with scaling variable $x = 12\nu c_v^2$ and crossover at $c_v \asymp (12\nu)^{-1/2}$: the deterministic limit $x \rightarrow 0$ recovers $(6\nu)^{1/3}$ and the jitter-dominated limit recovers

$2^{-1/3}c_v^{-2/3}$. An earlier $\min\{(6\nu)^{1/3}, c_0 c_v^{-2/3}\}$ conjecture is only the two-limit envelope of Eq. (5), with an artificial kink, and is superseded. Per the frozen claim discipline, the exact identity and the long-window limit *imply* the jitter exponent $-2/3$; the full finite-window crossover of Eq. (5) is a matched-asymptotic *prediction* until a uniform two-parameter (triangular-array renewal local-limit) proof is complete. The supplement’s crossover proposition is conditional on the stated uniform C^2 expansion; absent a separate proof of that expansion, the crossover remains a matched-asymptotic prediction.

The exponent pair $(1/3, -2/3)$ is universal within the regular finite-variance iid hidden-hold class: iid hidden nonnegative holds whose law is independent of λ , with finite nonzero mean and finite variance, forming a regular small-jitter family, under count-only observation and active-start nonparalyzable operation. It is *not* claimed for heavy-tailed (infinite-variance) holds, correlated or drifting holds, holds observed by the electronics, paralyzable detectors, event-time observations, or λ -dependent recovery. The precise critical statement is: zero hold-time variance is the critical manifold on which the optimal load continues to grow with window length; any fixed nonzero hidden hold variance produces a finite long-window optimum. Adjacent renewal-CLT and random-dead-time literature establishes the mean/variance ingredients and task-specific finite-rate optima [?, ?, ?, ?, ?]; we found no source stating the count-only exponent pair or the extensive/terminal crossover.

2.3 Preregistered verification

2.4 The conjugacy doctrine: photon budget versus time budget

The doctrine has two conjugate resource corners of the same dead-time information map. In the *photon-budgeted* corner the full-stack certificate holds the incident-signal resource as a hard *inequality*,

$$\int \rho_s(a) d\xi(a) \leq 0.60, \quad (6)$$

so a design measure may push a subset of high-value atoms into the ridge zone only if compensated by low-load or omitted atoms (R11 §4); detected counts are dual-reported but are never the sole fairness budget, because $\mathbb{E}[N \mid \rho] \simeq \nu\rho/(1+\rho)$ has vanishing marginal cost at high load. The *time-budgeted* corner is the no-gate ridge-tracking policy RIDGE-SCAT32: it fixes the balanced support and applies one global multiplier per dwell, calibrated on the pre-scan estimate to mean load $\rho_R(\nu)$, followed by a single global safety clip, spending a deliberately larger incident budget for time rather than photons (R11 §2). The exact production

ridge is $\rho_R(\nu) = \min \arg \max_{\rho \in \mathcal{R}_{\text{adm}}(\nu)} J_{\text{exact}}(\rho, \nu)$ over the admissible set $\mathcal{R}_{\text{adm}}(\nu) = \{0 < \rho \leq 24 : p_{\text{ceil}} \leq 0.01, J_{\text{mean}}/J_{\text{exact}} \geq 0.90\}$, using the exact numerical ridge rather than the asymptotic cube-root form (R11 §1). Under the incident-photon budget the relevant scalar efficiency is $J_{\text{exact}}(\rho, \nu)/\rho$, which decreases across the admitted palette; a photon-budgeted optimizer therefore rationally spends on lower loads and *avoids* the ridge, while the ridge belongs to the time-limited, power-available corner. This is a resource-regime statement, not a universal prohibition on ridge-zone allocation: sufficiently valuable directions may still receive larger loads in other cells (R13 §8(f)).

2.5 Why one global knob suffices for RIDGE-SCAT32

Fixed flux performs implicit information water filling when bucket brightness and local directional value satisfy the marginal-information alignment condition $\ell_i J'(\kappa u_i) = \beta$ across active directions ($\ell_i = q_i^\top V^{-1} q_i$ the directional value, $u_i = a_i^\top x$ the bucket brightness); its adequacy is measured by a projected KKT residual — not C_u alone — and outside that condition, variable-load OED can improve without any universal bound on the fixed-flux approximation ratio. On non-negative sparse-support scenes the source-induced load ordering naturally tracks directional value, which is why RIDGE-SCAT32 needs only a single global multiplier, and why the uniform per-pattern servo — which loses by up to ~ 7 dB in the companion’s dev evidence [?] — fails by erasing exactly that alignment. With random holds, J' vanishes at the jitter cap and is negative beyond it, so fixed flux can overdrive the brightest directions and explicit gain control becomes necessary. The full proposition, its residual near-optimality bound, and its failure modes are in the supplement. Naïve per-pattern load equalization was thus refuted (R11) for destroying this brightness–information alignment, while the development evidence of Section ?? shows the opposite extreme — adaptive concentration — colliding with the deployed conditioning stack; the balanced global-flux policy is the robust middle of the tested extremes, with dose-feasible geometry headroom outside the conditioning stack left to a separate follow-up (Section ??).

3 Preregistered campaign (M1)

M1 is a separate follow-up campaign with its own seed namespace, manifests, and expected-cell ledger; it does not reopen the companion’s frozen studies. It freezes on the R13 audit plus one immutable commit (tag `m1-freeze`); after that commit no scene, atom family, load grid, guard, endpoint, or gate may change, and if the primary fails there is no third redesign.

Geometry and constants. $n = 32^2 = 1024$; $M_{\text{total}} = 1024 = M_{\text{pre}}(52) + M_{\text{main}}(972)$ for every arm; nonparalyzable active start, $\tau = 50$ ns, $d = 0$, $\sigma_b = 0$; $\bar{\rho} \in \{0.05, 0.60\}$ for the budget-anchored arms; $\nu \in \{5, 10, 20, 50, 100, 200, 500, 1000, 2000\}$; five measurement seeds per (image, condition). Optical time $T_{\text{opt}} = M_{\text{total}} \nu \tau$; design-computation wall time is reported separately and never folded into T_{opt} .

Scenes. Twenty-four fresh confirmatory instances (six generator families $\times$ four instances) with seeds $s_{f,r} = 633000 + 100f + r$, plus six development-only instances $632900 + f$. The six committed generators (glyph, chirp, spokes, maze, contour, microtexture) are reused verbatim with their actual names — “USAF” is not used unless a USAF target is generated, and spokes is described as a radial resolution target. No acceptance filter of any kind is applied; generators, images, and ROIs receive SHA256 manifests.

Common pre-scan. Every arm receives and pays for the same 52-pattern scene-independent balanced pre-scan at its own $(\bar{\rho}, \nu)$: 32 paired 4×4 -block rows (support 32, amplitude 32), 16 block rows (amplitude 16), and 4 quadrant rows (amplitude 4), so every pixel receives cumulative amplitude $32 + 16 + 4 = 52$ and the row-mean pattern is exactly the all-ones vector; the mean pre-scan load then equals the declared global load for any sum-normalized scene, with no $\hat{x}$ -dependent row scaling (R13 §2). All 52 counts enter every arm’s final reconstruction; RIDGE-SCAT32 additionally uses the pre-scan estimate to calibrate its single global source multiplier, with no per-pattern servo. Every arm runs the same 972 SCAT32 main patterns and totals exactly 1024 physical exposures.

Arms and the deployed design. The deployed spatial design is the balanced exact 972-row SCAT32 multiset, identical across every operating-point comparison; the common 52-row pre-scan is charged to each arm. Table 1 lists the five arms. Three share this design and differ only in source power: SCAT32-SAFE at global mean load 0.05, SCAT32-060 at 0.60, and RIDGE-SCAT32, which applies one global multiplier per dwell (calibrated on the pre-scan to $\rho_R(\nu)$, then the frozen global clip). Two frozen fixed baselines, SCAT16 and LBLOB16 at load 0.60, are retained as descriptive context columns (no gate), so the cross-arm occupancy-ladder comparison keeps its reference columns. The $k = 32$ occupancy rung of the deployed design is fixed by the frozen development rule (median fixed-dwell radiometric PSNR at $(\bar{\rho}, \nu) = (0.60, 2000)$, averaged over development seeds), which selected SCAT32 over the clustered LBLOB16 and a matched baseline with development scores 17.498, 11.908, and 9.181 dB respec-

Table 1: The five M1 arms over the common balanced SCAT32 design. The paired RIDGE-SCAT32 vs. SCAT32-060 comparison carries the primary gate and RIDGE-SCAT32 vs. SCAT32-SAFE the speed comparison; the full-stack computation is descriptive and applies to the deployed design itself. SCAT16 and LBLOB16 are descriptive context.

Arm	Role	Gate
SCAT32-SAFE	balanced design at load 0.05 (safe reference; speed baseline)	speed base
SCAT32-060	balanced design at load 0.60 (primary comparator)	primary comp.
RIDGE-SCAT32	one global multiplier per dwell to $\rho_R(\nu)$ + global clip (ridge policy)	primary, speed
SCAT16	frozen scattered $k = 16$ at load 0.60 (companion bridge)	none
LBLOB16	committed clustered support at load 0.60 (spectral-floor reference)	none

tively; an earlier information-score-based selection was invalidated by external audit (R15) for departing from the frozen rule and was re-run under the original R10 prescription. The rung is consistent with the companion’s Study 2 fixed-dwell optimum [?]. Reconstruction (RQL with the frozen analytic λ_{TV} rule and pooled initialization) is identical across arms.

Endpoints and gates. Two preregistered empirical verdicts govern the campaign, alongside one descriptive structural analysis. The *primary* is a paired fixed-dwell ridge operating-point test (RIDGE_OPERATING_PASS). At terminal dwell $\nu = 2000$, for each confirmatory image j define $\Delta Q_j^{\text{ridge}} = \overline{\text{PSNR}}_{\text{rad}}(\text{RIDGE-SCAT32}, \nu=2000, j) - \overline{\text{PSNR}}_{\text{rad}}(\text{SCAT32-060}, \nu=2000, j)$, the mean over the five frozen seeds, with identical pre-scan, identical 972-row SCAT32 multiset, paired pattern/noise seeds, the same reconstruction path, and the same optical dwell — only the single global main-acquisition multiplier differs. PASS iff $\text{median}_j \Delta Q_j^{\text{ridge}} \geq 1.0$ dB, the 2.5% bootstrap lower bound on the median exceeds 0, and $\#\{j : \Delta Q_j^{\text{ridge}} > 0\} \geq 18/24$, using the existing 10,000-replicate nested family-stratified bootstrap; any numerical or guard failure stays in the intention-to-treat cohort and counts as nonpositive. This is a fixed-dwell, power-for-time operating-point test. It is not an equal-photon-budget or photon-efficiency comparison. The *secondary* is the nine-dwell Q_{90} time-to-quality test (RIDGE_SPEED_PASS): SCAT32-SAFE at load 0.05 against the dwell-dependent RIDGE-SCAT32 policy,

using the frozen PAVA/censoring machinery (the seed-mean radiometric PSNR, equal-weight PAVA isotonic curves, 0.50 dB safe-range informativeness floor, and censoring taxonomy transfer verbatim), with median $S_{\text{gate}} \geq 3$, bootstrap lower bound > 1 , and $\geq 18/24$ images with $S_{\text{gate}} > 1$. The *confirmatory structural secondary* is the full-stack descriptive analysis (Section ??). Every one of the $24 \times 5 \times 2 \times 2 = 480$ certificate cells ($\nu \in \{200, 2000\}$, $b \in \{0.05, 0.60\}$) receives exactly one terminal status — **CERTIFIED** (deployed design and class primal-feasible, all residual checks passed, $G_{\text{stack}}/r \leq 10^{-2}$), **COUNTEREXAMPLE** (a concrete full-stack-feasible design with actual log-determinant improvement per dimension $> 10^{-2}$, found by the frozen support-expanding primal screen), or **NUMERICAL_UNRESOLVED** (neither conclusion within the frozen numerical protocol) — under a frozen per-cell wall cap of 420 s (a 60 s primal screen, a 180 s cutting-plane solve, and one deterministic rescaled retry; no third attempt, no threshold change, no imputation, no cell deletion). The three statuses are descriptive: **COUNTEREXAMPLE** supplies constructive evidence of local geometry headroom, while **NUMERICAL_UNRESOLVED** supplies neither certification nor refutation. No categorical certificate verdict is assigned.

Disclosure ledger. Every cell discloses incident and detected photons (predicted and realized), the incident-dose and detected-count ratios, source/peak-irradiance ratio, the achieved mean load and its quantiles ρ_5 , ρ_{50} , ρ_{95} , ρ_{max} , the physical peak, predicted and realized ceiling fractions, any **RIDGE_GUARD_CLIPPED** flag, and mean J_{exact} ; certificate cells add the terminal status, the full-stack dual bound G_{stack} , the exact D-efficiency, solver status for the primary solve and the single retry, wall seconds, coefficient dynamic range, dictionary-scan and A-risk/spectral cut counts, the full-dictionary scan residual, the primal and dual gaps per r , the minimum generalized eigenvalue, the A-risk ratio, and the **MU_CAP_ACTIVE** disclosure. The full nine-dwell kernel/ridge certification table and the SHA256 of $D_{\text{cert}} = D_{\text{load}} \cup D_{\text{gain}}$ are frozen before any confirmatory scene is opened.

4 Results

4.1 Primary: ridge operating-point gain

The primary verdict is **RIDGE_OPERATING_PASS = PASS**. At $\nu = 2000$ the median paired radiometric gain of **RIDGE-SCAT32** over **SCAT32-060** is $\text{median}_j \Delta Q_j^{\text{ridge}} = 1.87$ dB (frozen bar ≥ 1.0 dB); the 2.5% nested family-stratified bootstrap lower bound on the median is 1.41348975 dB (bar > 0); and 19/24

Table 2: Per-family primary result at $\nu = 2000$: median paired ridge gain ΔQ^{ridge} and the count of positive scenes (four instances per family). The pooled median is 1.87 dB and 19/24 scenes are positive (**RIDGE_OPERATING_PASS**). Source: `results/round63_m1/M1.VERDICTS.json` and the imaging shards.

Family	median ΔQ^{ridge} (dB)	scenes > 0
maze	+6.37	4/4
glyph	+3.58	4/4
microtexture	+1.49	4/4
chirp	+1.40	4/4
spokes	+0.71	2/4
contour	−4.18	1/4

scenes are positive (bar $\geq 18/24$). All three conditions pass. The gain is bought with incident power, not photons: the ridge policy reaches its target mean load $\bar{\rho} = \rho_R(2000) = 22.25$ (the global safety clip does not bind), a $37\times$ incident-dose ratio and a $2.6\times$ detected-count ratio over the balanced **SCAT32-060** comparator, with no count-ceiling saturation (predicted $p_{\text{ceil}} < 10^{-6}$; realized dark-count and timing-error fractions 0) and no **RIDGE_GUARD_CLIPPED** event. The gain is strongly scene-dependent (Table 2, Figure 2), from +6.37 dB on **maze** to −4.18 dB on **contour**, the sole negative-median family (Section 5). This is a fixed-dwell, power-for-time operating-point test. It is not an equal-photon-budget or photon-efficiency comparison.

4.2 Secondary: ridge time-to-quality

Using the preregistered elapsed optical-time coordinate, $T_{\text{opt}} = M_{\text{total}}\nu\tau$, and the frozen Q90/PAVA/censoring procedure, the ridge time-to-quality endpoint passed: the median conservative speed ratio was $19.13\times$, the family-stratified nested-bootstrap 2.5th-percentile bound was $18.33\times$, and 21 of 24 scenes had $S_{\text{gate}} > 1$.

Analysis correction. After the campaign was unblinded, an independent peer audit found that the frozen scoring implementation did not reproduce three parts of the preregistered analysis. It placed the Q90 crossings on $\log(\nu\rho)$ rather than on the frozen elapsed optical time $\log T_{\text{opt}}$; it resampled family labels rather than retaining all six fixed strata with paired within-image seed resampling; and it used a nonstandard isotonic implementation while omitting the frozen censoring taxonomy. The immutable tag, raw cells, and original analyzer outputs were preserved unchanged. A dated correction artifact recomputed the endpoints from the same frozen raw cells using the preregistered axis and machinery; the promoted independent implementation and an in-project rerun agreed on all 18 audited numerical outputs with no deep differences. Scientific claims

and tables use the spec-conformant corrected results, while the original outputs remain available in the provenance record. The source-of-record bundle is `results/round63.m1/CORRECTION_2026-07-19/`, specifically `M1_VERDICTS_SPEC_CORRECTED_R19.json` and `ANALYSIS_CORRECTION_DISCLOSURE.md`.

As a post-hoc resource sensitivity, we repeated the corrected Q90 machinery on the incident-exposure coordinate proportional to $\nu\rho$. It yielded a median ratio of 0.279, a 2.5th-percentile bound of 0.240, and only 1 of 24 scenes above one. Thus ridge tracking greatly reduced elapsed dwell in this campaign but did not reduce incident exposure to Q90 under this coordinate. This sensitivity was not the preregistered speed endpoint and carries no PASS/FAIL verdict.

4.3 Structural secondary: descriptive full-stack analysis

Before the immutable campaign tag, the required 24-cell development feasibility screen selected the pre-registered `FALLBACK_DESCRIPTIVE` branch. Consequently, `FULL_STACK_CERT_PASS` was removed before freeze, and the post-freeze full-stack computation was specified only as a descriptive three-status analysis, not as a PASS/FAIL endpoint. Across the 480 frozen certificate cells (24 confirmatory scenes $\times$ 5 seeds $\times \nu \in \{200, 2000\} \times b \in \{0.05, 0.60\}$), 0 are `CERTIFIED`, 299 are `COUNTEREXAMPLE`, and 181 are `NUMERICAL_UNRESOLVED`; the fraction certified at $G_{\text{stack}}/r \leq 10^{-2}$ is 0. The status split is stable across the four anchors $((\nu, b) = (200, 0.05): 77/43; (200, 0.60): 67/53; (2000, 0.05): 84/36; (2000, 0.60): 71/49 \text{ counterexample/unresolved})$. Every cell is deployed-design feasible for budget, dose ($\pm 5\%$; realized deviation 0.040), A-risk, and spectral constraints, with no active finite dual cap (`MU_CAP_ACTIVE` false in all 480), and the protocol runs well inside its budget (median wall time 64.6 s, maximum 67.3 s, against the 420 s per-cell cap; no cell reaches it). The 299 counterexamples are concrete full-stack-feasible designs whose actual per-dimension log-determinant improvement over deployed SCAT32 exceeds the 10^{-2} bar by roughly two orders of magnitude (median 1.43, range 1.19–1.87 per dimension, found by the frozen support-expanding primal screen); the 181 unresolved cells returned an infeasible cutting-plane master under the frozen numerical protocol and are not certified — which is not the same as scientifically refuted. Under the pre-freeze descriptive branch had already been selected, this distribution is reported without a categorical certificate verdict. Descriptively, the counterexample cohort is confirmatory evidence that the balanced design’s scene-adapted geometry headroom survives the full deployed conditioning stack — now over fresh confirmatory scenes and the complete stack, consistent with the development-

only dose-only probe of Section ?? . Per-cell statuses, gaps, residuals, cut counts, and wall times are in the disclosure ledger (supplement S5.2).

5 Discussion: the operating handbook

The campaign exposes two conjugate resource regimes of the same dead-time information map. Under a linear incident-photon budget the per-photon efficiency $J_{\text{exact}}(\rho, \nu)/\rho$ falls monotonically, so a budget-constrained design rations photons onto lower loads and *avoids* the ridge; the ridge belongs to time-constrained operation, where power is available and the only scarcity is dwell. Reading the flux ridge and the design allocation as one program clarifies that the large fixed-dwell gains of the companion are an operating-point effect, not evidence that a design must spatially redistribute dose into the ridge.

The corrected verdicts are settled: the ridge operating-point primary *passes* (`RIDGE_OPERATING_PASS` = PASS, median 1.87 dB), the ridge speed secondary *passes* ($S_{\text{gate}} = 19.13\times$, lower bound $18.33\times$, 21/24 above one), and the full-stack analysis remains descriptive (0/480 cells certified); the campaign is preregistered to license only the governing ruling’s frozen conclusions. Global operating-point control remains the deployed positive arm; geometry selection is now known to retain dose-feasible headroom outside the conditioning stack (Section ??), so operating-point control and geometry adaptation are not presented as an exhaustive either-or. Because the descriptive branch was selected before freeze, the certificate licenses no categorical near-optimality statement and no collapse claim: across the 480 confirmatory cells, 0 certified, 299 counterexample, and 181 numerically unresolved (Section 4), so within the frozen full-stack class balanced SCAT32 is *not* certified near-optimal, and the descriptive counterexample cohort descriptively shows that scene-adapted geometry headroom survives the full deployed conditioning stack — a localized, better-defined open problem, not a rescued outcome. The certificate is comparator- and subspace-relative: it is stated relative to the frozen expanded dictionary, the local pre-scan estimate, the declared resource model, and the directions SCAT32 resolves. The line’s thesis, in the frozen wording of the governing ruling, is this. Uniform dose controls cumulative exposure but does not eliminate scene-adapted geometry selection. The operative restriction is the full deployed conditioning stack: balanced SCAT32 is evaluated descriptively within that frozen class, while time-limited image gains are tested separately by global ridge tracking.

Table 3: Cross-arm summary at the primary dwell $\nu = 2000$ (descriptive; means over 24 scenes $\times$ 5 seeds). Achieved $\bar{\rho}$ and incident-dose ratio are read from the imaging shards; the incident-dose ratio is normalized to the SCAT32-060 comparator. Ceiling fraction is the predicted count-saturation probability $p_{\text{ceil}} = P(N = \nu)$ from the frozen renewal kernel at the achieved mean load (realized dark-count fraction is 0 for every arm). Mean J_{exact} is the exact per-slot count information evaluated at the achieved mean load via the same frozen kernel. The gates are the RIDGE-SCAT32 primary and speed comparisons and the deployed-design certificate; “—” marks a paired metric that does not apply to a reference or context arm.

Arm	Role	achieved $\bar{\rho}$	inc.-dose ratio	ceiling frac.	mean J_{exact}	$\Delta Q^{\text{ridge}} / S_{\text{gate}}$
SCAT32-SAFE	safe ref. / speed base	0.05	0.08 $\times$	$< 10^{-6}$	0.048	—
SCAT32-060	primary comparator	0.60	1.00 $\times$	$< 10^{-6}$	0.375	—
RIDGE-SCAT32	ridge policy (gated)	22.25	37.1 $\times$	$< 10^{-6}$	0.935	+1.87 / 19.127
SCAT16	context	0.60	1.00 $\times$	$< 10^{-6}$	0.375	—
LBLOB16	context	0.60	1.01 $\times$	$< 10^{-6}$	0.375	—

The primary gain is scene-dependent, and its one negative-median family is diagnostic: on **contour** the balanced 0.60 design outperforms the ridge on three of four instances (median $\Delta Q^{\text{ridge}} = -4.18$ dB, versus +6.37 on **maze**), the separating covariate being bucket-brightness contrast (losing instances $C_u \in [0.21, 0.25]$ against the winning instance’s 0.33; across all 24 scenes ΔQ^{ridge} correlates with C_u at +0.64), with no guard, trust, dark-count, or ceiling flag firing, consistent with the alignment condition of Section 2.5 under which a single global flux multiplier is least effective where bucket brightness is weakly aligned with directional value; the residual family-geometry dependence — **contour** loses where an equally low-contrast **chirp** scene gains — is left as a hypothesis.

The jitter-capped ridge is the practical caution. The cube-root ridge is a property of the zero-jitter critical manifold; the moment a detector’s hold time carries a coefficient of variation c_v , the long-window optimum is finite — the exact cubic $c_v^2 \rho^2 (1 + 2\rho) = 1$ of Theorem 3, scaling as $2^{-1/3} c_v^{-2/3}$ — and driving past it is actively harmful: at the deterministic ridge ($\rho = 22.3$), 5% jitter already forfeits about 55% of the scalar count information in the refined sweep (Table ??). A deployment must therefore locate its operating point against the jitter-limited ceiling of the real detector, not the deterministic ridge; the safe conclusion is conditional on measured timing jitter.

Together, the law and the design principle compress into the operating handbook of Table 4: characterize the detector, let the law set the load ceiling, check engagement with the pre-scan, and pick the pattern recipe by which resource is scarce.

Two scope boundaries stand. All results are simulation-based, and every certificate is stated relative to the frozen dictionary, the local pre-scan estimate, and the declared resource model; an optical extension along the companion’s discussion — a coarse complex-field estimate synthesizing an optical displacement that removes common-mode photons before dead-time loss — remains a natural bench direction [?].

Table 4: Operating handbook for dead-time-limited single-pixel imaging. The engagement index Γ is descriptive (the companion’s computable engagement criterion [?]), never a confirmatory gate.

Step	Rule
1. Characterize	measure τ and the hold-time c_v from time-tagged recovery intervals
2. Set the load	jitter-capped optimum from $c_v^2 \rho^2 (1 + 2\rho) = 1$ (long window) or $(6\nu)^{1/3} (1 + 12\nu c_v^2)^{-1/3}$ (finite window); never operate a jittered detector at the deterministic ridge
3. Check engagement	common 52-pattern pre-scan $\rightarrow$ measured C_u ; $\Gamma = C_u \sqrt{\nu \rho / (1 + \rho)} \gtrsim 1$ marks the photon-limited regime
4. Pick patterns	photon-budgeted: the balanced SCAT32 design, tested for certified near-optimality within the full deployed conditioning class ($G_{\text{stack}}/r \leq 10^{-2}$); time-budgeted: RIDGE-SCAT32 — one global multiplier to $\rho_R(\nu)$ (Section 2.5)
5. Doctrine	photon-constrained $\rightarrow$ stay below the ridge (J_{exact}/ρ falls); time-constrained $\rightarrow$ track the jitter-capped ridge

The law itself, however, is bench-testable now, with no imaging required. For a real single-photon counting module, the hold-time coefficient of variation is directly measurable from time-tagged recovery intervals; the theory then predicts a count-information peak at the root of $c_v^2 \rho^2 (1 + 2\rho) = 1$ rather than at the deterministic ridge — for $c_v = 0.05$, a peak near $\rho \approx 5.7$ instead of ≈ 22 at $\nu = 2000$, with about half the information forfeited if the deterministic ridge is used instead (Table ??). Sweeping source power at fixed dwell and estimating the count information from repeated frames traces $J(\rho)$ directly; this single-detector power sweep is the immediate bench experiment for the line.

6 Reproducibility

The design, guards, and campaign follow the frozen rulings R10–R18: exact renewal-kernel OED, incident-

budget conjugacy, the M1 audit and unification, the safety-constrained refreeze, and full-stack localization. Together they retired the adaptive imaging arm, retained the dose-only probe as development-only evidence, and fixed the corrected claim discipline used here. Exact-law peak predictions were preregistered before the refined jitter sweep and confirmed by grid, local-zoom, and pooled replication audits; the intermediate $k = 4/3$ hypothesis was refuted. The ruling documents and commit identifiers are the audit trail. The campaign freezes at tag `m1-freeze` once the R13 checklist passes outcome-blind self-tests; the companion is frozen at tag `paper1-freeze-v1`. Source-of-record documents for every frozen constant are cited by ruling number in the block comments of the manuscript source; full machinery and proofs are in the supplement (`supplement.m1.tex`). All code, frozen pre-registration documents, per-cell manifests, and evidence bundles are available at [\[repo URL wording — user decision\]](#); funding and acknowledgments are [\[funding/acknowledgments — user decision\]](#).

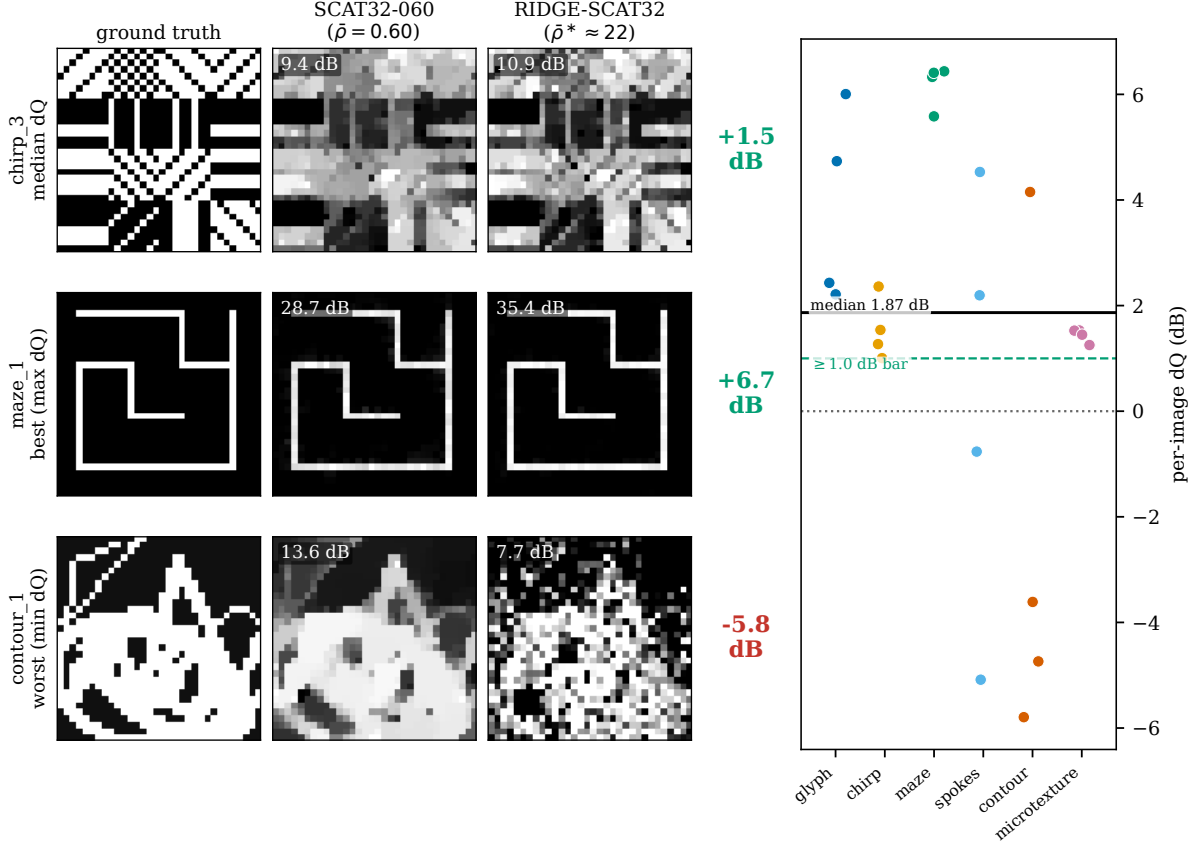


Figure 2: Fixed-dwell reconstruction quality of the global power knob: RIDGE-SCAT32 against balanced SCAT32-060 at $\nu = 2000$, seed 0 (the centrepiece of the operating-point primary). Each row shows, left to right, the ground truth, the SCAT32-060 reconstruction ($\bar{\rho} = 0.60$), and the RIDGE-SCAT32 reconstruction ($\bar{\rho}^* \approx 22$), annotated with the per-row ΔQ badge: the median- ΔQ scene m1_chirp_3 (radiometric PSNR_{rad} of 9.4 and 10.9 dB, respectively; $\Delta Q = +1.5$ dB), the best- ΔQ scene m1_maze_1 (28.7 and 35.4 dB; $\Delta Q = +6.7$ dB), and the worst- ΔQ scene m1_contour_1, a contour (13.6 and 7.7 dB; $\Delta Q = -5.8$ dB). The right panel plots the per-image ΔQ (seed-mean over five seeds) for all 24 confirmatory scenes grouped by family; the solid line is the campaign median 1.87 dB, the dashed line the 1.0 dB operating bar, and 19/24 scenes exceed zero. PSNR_{rad} denotes the radiometric peak signal-to-noise ratio and $\Delta Q = \text{PSNR}_{\text{rad}}(\text{RIDGE-SCAT32}) - \text{PSNR}_{\text{rad}}(\text{SCAT32-060})$; the thumbnail values are the seed-0 realization. The colour is used only for display purposes, and each row shares a common grayscale scale.

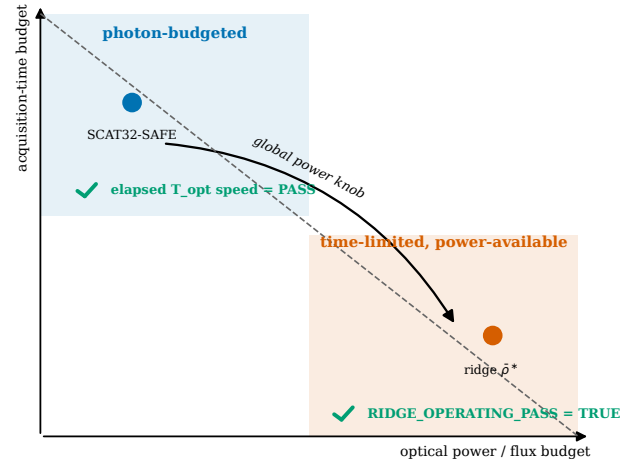
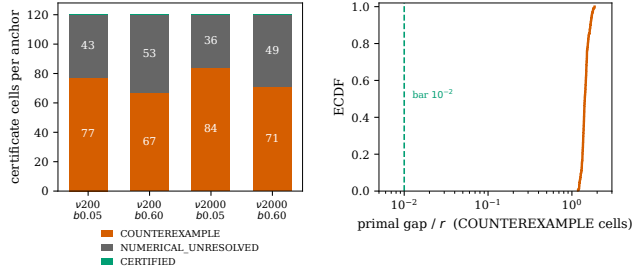
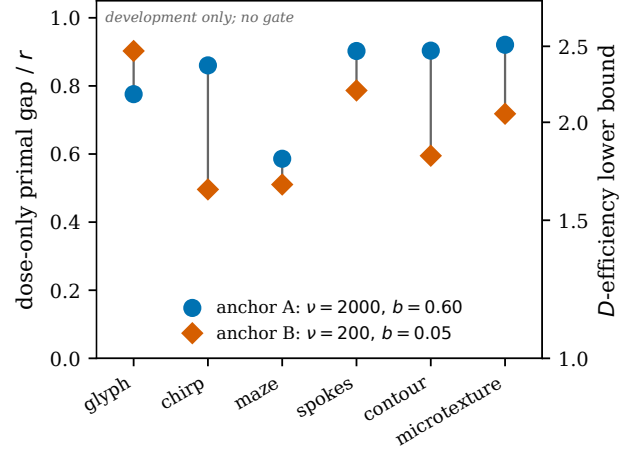
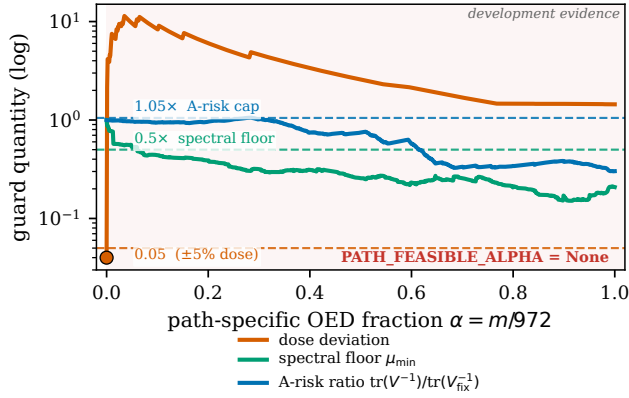


Figure 3: Act III design doctrine: information rewards scene-adapted geometry, uniform dose alone does not eliminate that headroom (development evidence, panels a–b), the conditioning stack defines the deployable class tested by the full-stack certificate (panel c), and global operating-point control survives orthogonally (panel e; the operating-point reconstructions are Figure 2). (a) Guard collapse of the adaptive OED-DT mixture along the descending exact-rounding path for the development glyph scene at the fast anchor ($\nu = 2000$, $b = 0.60$); development-only evidence, not a confirmatory endpoint. The dose deviation $\max |d/\bar{d} - 1|$ (orange), the spectral floor $\mu_{\min}$ (green), and the A-risk ratio $\text{tr}(V^{-1})/\text{tr}(V_{\text{fix}}^{-1})$ (blue) are shown versus the path-specific OED fraction $\alpha = m/972$, with the dashed guard bars the $\pm 5\%$ dose band (0.05), the spectral floor ($0.5\times$), and the A-risk cap ($1.05\times$), respectively; every $\alpha > 0$ exceeds the dose band by $2\text{--}11\times$ and, by $\alpha \approx 0.06$, breaches the spectral floor, while the A-risk cap never binds and only the balanced endpoint $\alpha = 0$ is dose-feasible (failing the separate efficiency floor $\text{eff}_D \geq 0.95$), so **PATH_FEASIBLE_ALPHA** is **NONE** (**NO_FEASIBLE_POINT_ON_PATH**). (b) Constructive dose-only geometry headroom over the six development families; development-only lower bound, not a confirmatory endpoint or population estimate. The per-dimension dose-only primal gap $[\log \det V(\xi) - \log \det V_{\text{SCAT32}}]/r$ within the $\pm 5\%$ dose band is shown at anchor A ($\nu = 2000$, $b = 0.60$; circles) and anchor B ($\nu = 200$, $b = 0.05$; diamonds), the right axis reporting the D-efficiency lower bound $\exp(\text{gap})$; across glyph, chirp, maze, spokes, contour, and microtexture the gaps are 0.78, 0.86, 0.59, 0.90, 0.90, and 0.92 at anchor A and 0.90, 0.50, 0.51, 0.79, 0.59, and 0.72 at anchor B, respectively (D-efficiency lower bounds $1.6\text{--}2.5\times$). The construction is support-preserving and establishes existence, not the dose-only optimum. (c) Descriptive full-stack analysis distribution under the fallback-descriptive branch (0/480 **CERTIFIED**). The left bars give the terminal status of the 480 certificate cells split by the four (ν, b) anchors (120 cells each): at $(\nu 200, b 0.05)$, $(\nu 200, b 0.60)$, $(\nu 2000, b 0.05)$, and $(\nu 2000, b 0.60)$ the cells are 77, 67, 84, and 71 **COUNTEREXAMPLE** and 43, 53, 36, and 49 **NUMERICAL_UNRESOLVED**, respectively, with no **CERTIFIED** cell; the right panel is the empirical cumulative distribution of the finite primal gap per dimension for the 299 **COUNTEREXAMPLE** cells (median 1.43), the dashed line the near-optimality bar $G_{\text{stack}}/r \leq 10^{-2}$, which every counterexample exceeds by about two orders of magnitude. The certificate is reported descriptively and no categorical near-optimality claim is made. (e) Conjugate-corners operating doctrine for the global power knob: the plane is spanned by the optical-power/flux budget (horizontal) and the acquisition-time budget (vertical); in the photon-budgeted corner (low power, ample time; Q90 elapsed-time secondary) the SCAT32-SAFE comparator is used for the corrected preregistered speed endpoint, whereas in the time-limited, power available corner (ample power, little time; fixed dull quality primary) the ridge operating point $\bar{\delta}^*$ delivers